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/*
 * C Program to Implement a Queue using an Array
 */
#include <stdio.h>

#define MAX 5

void insert();
void delete();
void display();

int queue_array[MAX];

int rear = - 1;

int front = - 1;

main()
{
    int choice;
    while (1)
    {
        printf("1.Insert element to queue \n");
        printf("2.Delete element from queue \n");
        printf("3.Display all elements of queue \n");
        printf("4.Quit \n");
        printf("Enter your choice : ");
        scanf("%d", &choice);
        switch (choice)
        {
            case 1:
                insert();
                break;
            case 2:
                delete();
                break;
            case 3:

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        display();

        break;

        case 4:

            exit(1);

        default:

            printf("Wrong choice \n");

    } /* End of switch */

} /* End of while */

} /* End of main() */

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void insert()
{
    int add_item;

    if (rear == MAX - 1)
        printf("Queue Overflow \n");
    else
    {
        if (front == - 1)

            /*If queue is initially empty */

            front = 0;

        printf("Inset the element in queue : ");

        scanf("%d", &add_item);

        rear = rear + 1;

        queue_array[rear] = add_item;
    }

} /* End of insert() */

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void delete()
{
    if (front == - 1 || front > rear)

    {

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    printf("Queue Underflow \n");
    return ;
}
else
{
    printf("Element deleted from queue is : %d\n", queue_array[front]);
    front = front + 1;
}
} /* End of delete() */
```

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void display()
{
    int i;
    if (front == - 1)
        printf("Queue is empty \n");
    else
    {
        printf("Queue is : \n");
        for (i = front; i <= rear; i++)
            printf("%d ", queue_array[i]);
        printf("\n");
    }
} /* End of display() */
```